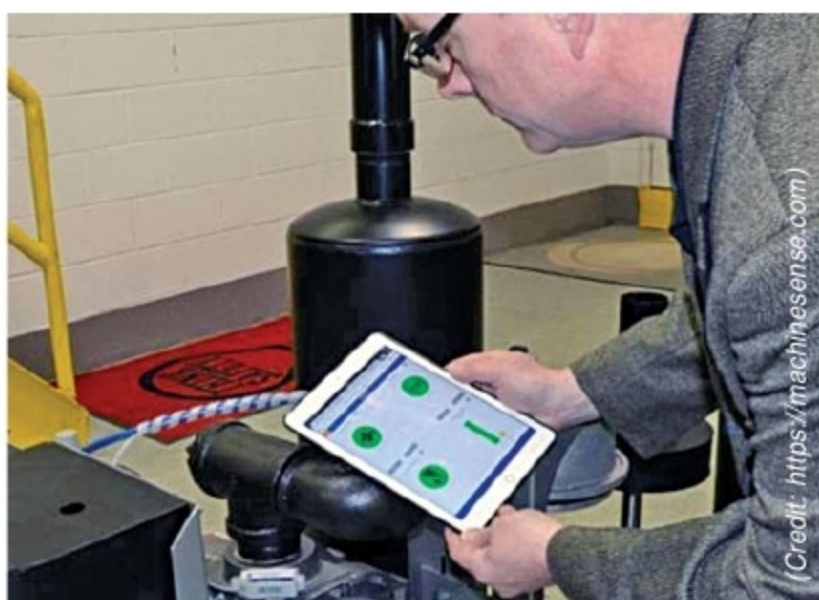


machines for issues in a bearing, blower, or a broken tooth in gearbox. Electrical characteristics are used to analyse electrical sub-systems to understand specific power quality anomalies within industrial and commercial power settings.

In this technology, the MEMS-based wearable sensors can be easily attached to the outside of the machine or components to automatically monitor their condition on a regular basis. The sensor data is transmitted to an easy-to-install gateway, from where it is further sent to a cloud-based server running powerful analytic software. The server transmits results to a user-friendly app, where one can view real-time machine condition to give maintenance advice when required.

The software packages installed across all industrial machinery pro-



vide a status dashboard of all connected assets such as analytics, histograms, and maintenance, repairs and operational advice. Local machine intelligence is allocated to mobile apps to provide gauge visualisations and maintenance corrective action alerts.

The Wi-Fi- and Ethernet-enabled data hub can connect with multiple sensors that enable edge computing for at-site storage and analytics. Its

advanced meta-data analytics engine provides deep machine learning and statistical analysis. It never records process data, and never takes control of the machine or components, to avoid potential security threats.

Along with strong analytics and trusted data sources, there are also industrial IoT (IIoT) tools for downtime reduction of sensors, edge devices, or sensor connectivity.

The scalability of the product or system becomes an issue when the number of sensors and devices exceeds, say, 250 in an IIoT system. As every device has to be brought up to the same firmware or middleware version, connectivity becomes a major challenge in that case. To mitigate these challenges, one has to update the algorithms in the device as well, using OTT (over the top) architecture.

—Deepshikha Shukla

4 LEVERAGING TRAPPED-ION TECHNOLOGY FOR QUANTUM COMPUTING

Quantum computing is considered an efficient and precise way of solving complex problems which may not get easily solved by even the largest supercomputers in fields like chemistry, medicine, finance, and logistics. In comparison to the rest of the modern computing methods, it consumes less power and speed of computation is remarkably high.

With the aim of taking trapped-ion quantum computing out of the lab and into the market, US-based startup IONQ was founded in 2015 by Chris Monroe and Jungsang Kim. They initially received \$2 million in seed funding from New Enterprise Associate and a licence to core technology from the University of Maryland and Duke University necessary for developing trapped-ion quantum computers and software.

A quantum computer takes advantage of properties of

Bits vs Qubits

In a classical computer, the smallest unit of computation is called a bit. A bit can be in only one of two possible states: one or zero. A qubit has the ability to be in a complex superposition of these states, allowing for a much larger computational space—imagine only being able to point at the north or south pole, versus being able to point anywhere on the globe.

quantum systems such as entanglement and superposition to perform mathematical calculations. Instead of using synthetic quantum systems, the startup uses individual atoms for their quantum processing units to ensure reliable and scalable quantum computing. The team chose atoms of ytterbium, a silvery rare-earth metal, as these are perfectly identical to each other and, once prepared, they can remain in a particular stable quantum state for very long periods of time.

An electron from a neutral ytterbium atom is removed with lasers, resulting in a ytterbium ion from this trapping process. Ionised ytterbium atoms then act as quantum bits (qubits), the smallest unit of computation. These ions are held in 3D space using electromagnetic forces produced from a specialised chip called a linear ion trap that consists of around hundred tiny electrodes with rapidly oscillating voltages.

